

SYED MASRUR AHMED

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EDUCATION

Dartmouth College

Hanover, NH

Bachelor of Arts, Major in Mathematics and Computer Science, GPA 3.94/4.00

June 2028

- **Relevant Coursework:** Data Structures & Algorithms, Discrete Math, Fullstack Development, Computer Architecture, Machine Learning and Statistical Methods.
- **Clubs and Organizations:** Dartmouth Quantitative Finance Club, Dartmouth Tech Consulting Group, Dartmouth Student Blockchain, Great Issues Scholars

SKILLS

Programming Languages: Python, C, C++, Assembly, Java, JavaScript, C#, SQL, Bash, HTML, CSS

Technologies: Git, Docker, Next.js, Flask, FastAPI, PyTorch, Pandas, Unity, AWS (Lambda, S3), Agentic AI

WORK EXPERIENCE

The National Law Review

Remote

Software Engineering Intern

June 2026 – Present

- Architected end-to-end Python web scraping workflows to aggregate unstructured data from diverse municipal websites, powering a centralized public notices database proposed to the New Hampshire governing body.
- Engineered full-stack platform features using custom Drupal PHP modules and responsive frontend interfaces, automating ETL migrations while implementing robust anti-scraping system architecture.

Department of Mathematics, Dartmouth College

Hanover, NH

Machine Learning Research Assistant

March 2025 – Present

- Developed a custom transformer architecture in PyTorch utilizing Hermite-polynomial input embeddings to model continuous stochastic processes, outperforming standard linear projections in Gaussian-distributed environments.
- Engineered an end-to-end research pipeline that mathematically interpolates between discrete Markov chains and continuous Ornstein-Uhlenbeck processes to stress-test in-context learning thresholds.

DALI Lab (Dartmouth Applied Learning & Innovation)

Hanover, NH

Software Engineer (Part-time)

September 2025 – March 2026

- **SimReach (Healthcare SaaS):** Engineered an AWS-based decision support tool to optimize physician outreach, architecting a data validation algorithm for EHR uploads and building full-stack onboarding flows with automated Lambda-driven cron jobs.
- **Benchmark Space Systems:** Scaled a Godot satellite simulation to 1,000+ users via a web port and a serverless Firebase backend.

PROJECTS

MaskProxy | [Github](#) | *Rust, Cloudflare Pingora, Redis, LanceDB, Next.js*

March 2026 – April 2026

- Engineered a high-performance privacy middleware in Rust using Cloudflare Pingora to intercept LLM API calls, utilizing NER models (GLiNER/DeBERTa-v3) to detect and mask PII with deterministic tokens stored in Redis for stateful rehydration.
- Architected a semantic routing engine with LanceDB and BGE-Small embeddings to autonomously direct sensitive prompts to local Phi-3.5 instances, while developing a Next.js dashboard for real-time observability into masking and routing decisions

Trippee | [trippee-ai.vercel.app](#) | [Github](#) | *Next.js 15, TypeScript, Supabase, Google Gemini, Mapbox*

December 2025

- Engineered a real-time collaborative trip-planning platform using Next.js 15 and Supabase, utilizing WebSockets to synchronize geospatial map state, live user presence, and shared rich-text editing across client sessions.
- Architected an agentic AI workflow using Google Gemini for autonomous database interactions via tool calling, and applied K-Means clustering algorithms to automate itinerary structuring and route optimization.

HONORS AND AWARDS

- **Academic Distinctions:** Rufus Choate Scholar (Awarded to the top 5% of students), Jack Byrne Scholar in Mathematics; Academic Citations for Excellence in Linear Algebra and Machine Learning.
- **International Achievements:** Highest Score in the World (A-levels Chemistry); Highest Score in Bangladesh (A-level Further Mathematics)